

Muhammad Kashif, Ph.D.

Postdoctoral Research Associate, Center for Quantum and Topological Systems
Research Team Lead, eBRAIN Lab

New York University Abu Dhabi, United Arab Emirates

muhammadkashif@nyu.edu | muhammadkashif038@gmail.com

Research website | Google Scholar | ORCID: 0000-0003-2023-6371 | LinkedIn

Citation metrics: 1,150+ citations; *h-index* 21; *i10-index* 33 (October 2026)

Research Profile

My research connects quantum algorithms, machine learning, and hardware-aware quantum computing. I develop and benchmark hybrid quantum–classical learning methods, investigate trainability and noise robustness, and design resource-aware architectures using analytical and hardware-calibrated cost models. My recent work also explores generative AI for quantum programming and quantum optimization for practical applications.

Research interests: Quantum machine learning; variational quantum algorithms; trainability and barren plateau mitigation; quantum resource estimation; hardware-aware neural architecture search; quantum advantage assessment; generative AI for quantum software.

Professional Experience

Postdoctoral Research Associate

Aug 2023–Present

Center for Quantum and Topological Systems, New York University Abu Dhabi, UAE

Additional current role: Research Team Lead, eBRAIN Lab.

- Develop analytical and hardware-calibrated quantum resource cost models and resource-aware neural architecture search methods that evaluate accuracy–cost trade-offs.
- Analyze the effects of noise, connectivity, and compilation on quantum circuit performance, scalability, expressibility, and trainability.
- Develop noise-resilient training and barren plateau mitigation approaches for feedforward, quantum convolutional, and graph recurrent quantum neural networks.
- Investigate quantum optimization methods, including QAOA and VQE, and quantum workflows for path planning, unit commitment, and robot navigation.
- Develop domain-specific large language models, datasets, and benchmarks for automated quantum code generation with PennyLane.
- Benchmark hybrid quantum–classical models for financial prediction, classification, and healthcare applications; supervise undergraduate and graduate research projects.

Research and Teaching Assistant

Mar 2018–Dec 2019

Istanbul Şehir University, Turkey

- Conducted undergraduate laboratory and physics problem-solving sessions for 60+ students.
- Designed, optimized, and evaluated FPGA-based elliptic curve cryptographic accelerators.

Research Engineer (Remote)

Feb 2015–Aug 2017

Collaborative research projects, National Science, Technology and Innovation Plan (NSTIP), Saudi Arabia

- Developed structural and behavioral SysML/MARTE models of safety-critical embedded systems, including unmanned aerial vehicles, intelligent traffic controllers, and collision avoidance systems.
- Implemented FPGA-based elliptic curve crypto-processors and optimized area and throughput.
- Implemented sorting algorithms in C for MATLAB-based genetic algorithm workflows supporting temporal verification and execution-time prediction.

Education

- Ph.D., Computer Science and Engineering** 2020–2023
Hamad Bin Khalifa University, Qatar
Thesis: *Quantum Machine Learning on Noisy Intermediate-Scale Quantum Devices.*
- M.Sc., Electronics and Computer Engineering** 2018–2019
Istanbul Şehir University, Turkey
Thesis: *A Hardware-Efficient Elliptic Curve Accelerator for FPGA-Based Cryptographic Applications.*
- B.Sc., Electrical (Electronics) Engineering** 2011–2015
COMSATS, Pakistan
Final-year project: *Home Automation System Using Raspberry Pi over Wi-Fi.*

Teaching, Supervision, and Mentoring

Teaching and Educational Materials

- **Undergraduate teaching assistance, Istanbul Şehir University (2018–2019):** Laboratory instruction and physics problem-solving sessions for 60+ students.
- **Model-based systems engineering materials, NSTIP (2015–2017):** Prepared tutorials and case studies that supported course development.

Thesis Co-Supervision

- **M.Sc., Université Libre de Bruxelles (2024–2025):** *Noise Mitigation Techniques for Reliable Quantum Neural Networks.*
- **B.Sc., New York University Abu Dhabi (2023–2024):** *Comparative Analysis of Noise and Robustness Evaluation in Quantum Neural Networks.*

Research and Hackathon Mentoring

- Supervise undergraduate and graduate research projects at NYU Abu Dhabi in quantum learning, optimization, and quantum software.
- [Mentor, NYUAD Quantum Hackathon for Social Good](#), 2024, 2025, and 2026.
- [Mentor, Quantum Computing Hackathon Pakistan](#), 2026; mentored team placed second runner-up.

Teaching Interests

Quantum computing; quantum machine learning; machine learning and AI; algorithms and optimization; digital logic and systems design; computer engineering; embedded systems and hardware design.

Awards and Honors

- **Best Paper Nominee, IJCNN 2026:** *Design Space Exploration of Hybrid Quantum Neural Networks for Chronic Kidney Disease.*
- **Best Paper Nominee, IJCNN 2026:** *A PennyLane-Centric Dataset to Enhance LLM-Based Quantum Code Generation Using RAG.*
- **First Prize, QInnovation World Challenge**, Thales, Quantum Innovation Summit, Dubai, UAE, 2025. [Challenge](#)
- **Best Researcher Award**, eBRAIN Lab, NYU Abu Dhabi, 2024–2025.
- **Fully funded Ph.D. scholarship**, Hamad Bin Khalifa University, 2020–2023.
- **Fully funded M.Sc. scholarship (TA/RA)**, Istanbul Şehir University, 2018–2019.
- **Outstanding Performance Award**, NSTIP, Saudi Arabia, 2015–2017.

Professional Service and Leadership

- **Research leadership:** Research Team Lead, eBRAIN Lab, NYU Abu Dhabi; coordinate collaborative research and student mentoring.
- **Technical program committee:** IEEE International Workshop on Quantum Computing: Circuits, Systems, Automation, and Applications (QC-CSAA), co-located with ISVLSI 2026.
- **Workshop and special-session organization:** Co-organized sessions and workshops at IEEE IJCNN and IEEE Quantum Week (QCE).
- **Challenge co-organization:** [Quantum Federated Learning for Fraud Detection](#), Quantum Innovation Summit, Dubai, 2025.
- **Journal reviewing:** IEEE Transactions on Emerging Topics in Computing; IEEE Transactions on Quantum Engineering; IEEE Transactions on Computer-Aided Design; Advanced Quantum Technologies; Engineering Applications of Artificial Intelligence; Computers & Electrical Engineering; Expert Systems with Applications; Journal of Computational Physics; Neurocomputing; SoftwareX.
- **Conference reviewing:** IEEE International Conference on Quantum Computing and Engineering (QCE); International Joint Conference on Neural Networks (IJCNN), Design Automation and Test in Europe (DATE)

Conference Presentations

- **DAC 2026**, Long Beach, California, USA, July 2026: *Closing the Loop: Resource-Aware Hybrid NAS Guided by Analytical and Hardware-Calibrated Quantum Cost Modeling*.
- **DAC 2025**, San Francisco, California, USA, June 2025: *Computational Advantage in Hybrid Quantum Neural Networks: Myth or Reality?*
- **IEEE Quantum Week (QCE) 2025**, New Mexico, USA, September 2025: *Multiple papers*
- **ICRC 2024**, San Diego, California, USA, December 2024: *The Dilemma of Random Parameter Initialization and Barren Plateaus in Variational Quantum Algorithms*.

Technical Skills

- **Quantum computing:** PennyLane, Qiskit, IBM Quantum backends; hybrid quantum neural networks; noise modeling; robustness and trainability analysis.
- **Machine learning and scientific computing:** Python, NumPy, SciPy, scikit-learn, Keras, TensorFlow; experimental benchmarking.
- **Optimization and system evaluation:** Resource-aware neural architecture search; analytical and hardware-calibrated cost modeling; circuit-depth and two-qubit gate analysis.
- **Hardware and embedded systems:** C/C++, MATLAB, Verilog, VHDL, FPGA, Vivado, SysML, MARTE.

Publications

Published papers, accepted conference papers, and preprints are listed separately. Preprint versions of accepted or published papers are linked within the corresponding entry rather than listed twice. The author's name is shown in bold.

Three Representative Publications

- **ResQNETs** (EPJ Quantum Technology, 2024): Residual learning for mitigating barren plateaus. [Publication](#)
- **Computational Advantage in Hybrid Quantum Neural Networks: Myth or Reality?** (DAC 2025). [Publication](#)
- **Closing the Loop** (DAC 2026, accepted and presented; proceedings pending). [Manuscript](#)

Journal Articles

- J1 T. Ahmed, **M. Kashif**, A. Marchisio, and M. Shafique. “A comparative analysis and noise robustness evaluation in quantum neural networks.” *Scientific Reports*, 2025. [DOI](#)
- J2 **M. Kashif** and M. Shafique. “Deep quanvolutional neural networks with enhanced trainability and gradient propagation.” *Scientific Reports*, 2025. [DOI](#)
- J3 **M. Kashif** and S. Al-Kuwari. “ResQNNs: A residual approach for mitigating barren plateaus in quantum neural networks.” *EPJ Quantum Technology*, 2024. [DOI](#)
- J4 **M. Kashif** and S. Al-Kuwari. “Physical realization of measurement based quantum computation.” *IEEE Access*, 2023. [Publisher](#)
- J5 **M. Kashif** and S. Al-Kuwari. “The impact of cost function globality and locality in hybrid quantum neural networks on NISQ devices.” *Machine Learning: Science and Technology*, 2023. [DOI](#)
- J6 **M. Kashif** and S. Al-Kuwari. “The unified effect of data encoding, ansatz expressibility and entanglement on the trainability of HQNNs.” *International Journal of Parallel, Emergent and Distributed Systems*, 2023. [DOI](#)
- J7 M. Rashid, M. Imran, **M. Kashif**, and A. Sajid. “An optimized architecture for binary Huff curves with improved security.” *IEEE Access*, 2021. [DOI](#)
- J8 **M. Kashif** and I. Çiçek. “Design space exploration of hybrid quantum–classical neural networks.” *Electronics*, 10(23), 2980, 2021. [Publisher](#)
- J9 **M. Kashif** and I. Çiçek. “Field-programmable gate array (FPGA) hardware design and implementation of a new area efficient elliptic curve crypto-processor.” *Turkish Journal of Electrical Engineering and Computer Sciences*, 29(4), 2127–2139, 2021. [DOI](#)
- J10 M. W. Anwar, M. Rashid, F. Azam, A. Naeem, **M. Kashif**, and W. H. Butt. “A unified model-based framework for the simplified execution of static and dynamic assertion-based verification.” *IEEE Access*, 2020. [DOI](#)
- J11 M. Rashid, S. A. B. Shah, M. Arif, and **M. Kashif**. “Determination of worst-case data using an adaptive surrogate model for real-time system.” *Journal of Circuits, Systems and Computers*, 2020. [DOI](#)
- J12 M. W. Anwar, M. Rashid, F. Azam, **M. Kashif**, and W. H. Butt. “A model-driven framework for design and verification of embedded systems through SystemVerilog.” *Design Automation for Embedded Systems*, 2019. [DOI](#)
- J13 M. Imran, M. Rashid, A. R. Jafri, and **M. Kashif**. “Throughput/area optimised pipelined architecture for elliptic curve crypto processor.” *IET Computers & Digital Techniques*, 2019. [DOI](#)
- J14 M. W. Anwar, M. Rashid, F. Azam, and **M. Kashif**. “Model-based design verification for embedded systems through SVOCL: An OCL extension for SystemVerilog.” *Design Automation for Embedded Systems*, 2017. [DOI](#)

Accepted Conference Papers — Manuscript Links

- A1 **M. Kashif**, A. Marchisio, and M. Shafique. “Closing the Loop: Resource-aware Hybrid NAS Guided by Analytical and Hardware-Calibrated Quantum Cost Modeling.” *ACM/IEEE Design Automation Conference (DAC)*, 2026; accepted and presented; proceedings pending. [arXiv:2603.00625](#)
- A2 **M. Kashif** and M. Shafique. “Late Breaking Results: Hardware-aware compilation reshapes trainability in variational quantum circuits.” *ACM/IEEE Design Automation Conference (DAC)*, 2026; accepted. [arXiv:2604.16527](#)

- A3 **M. Kashif**, S. Khalid, A. Marchisio, N. Innan, and M. Shafique. “FAQNAS: FLOPs-aware hybrid quantum neural architecture search using genetic algorithm.” *Design, Automation & Test in Europe (DATE)*, 2026; accepted. [arXiv:2511.10062](#)
- A4 **M. Kashif**, H. M. Siraj, N. Innan, A. Marchisio, and M. Shafique. “Design space exploration of hybrid quantum neural networks for chronic kidney disease.” *International Joint Conference on Neural Networks (IJCNN)*, 2026; accepted; Best Paper Nominee. [arXiv:2604.13608](#)
- A5 T. Ahmed, A. Marchisio, **M. Kashif**, N. Innan, and M. Shafique. “GAT-QNN: Genetic algorithm-based training of hybrid quantum neural networks.” *International Joint Conference on Neural Networks (IJCNN)*, 2026; accepted. [arXiv:2604.15048](#)
- A6 A. Basit, N. Innan, M. H. Asif, M. Shao, **M. Kashif**, A. Marchisio, and M. Shafique. “A PennyLane-Centric Dataset to Enhance LLM-based Quantum Code Generation using RAG.” *International Joint Conference on Neural Networks (IJCNN)*, 2026; accepted; Best Paper Nominee. [arXiv:2503.02497](#)
- A7 M. Shao, N. Innan, H. Janardhanan, **M. Kashif**, A. Marchisio, and M. Shafique. “PennySynth: RAG-driven data synthesis for automated quantum code generation.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026; accepted. [arXiv:2605.25572](#)
- A8 M. K. Altrabulsi, N. Innan, A. Marchisio, **M. Kashif**, and M. Shafique. “Q-SpiRL: Quantum spiking reinforcement learning for adaptive robot navigation.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026; accepted. [arXiv:2605.20801](#)

Published Conference and Proceedings Papers

- C1 **M. Kashif**, A. Marchisio, and M. Shafique. “Computational advantage in hybrid quantum neural networks: Myth or reality?.” *ACM/IEEE Design Automation Conference (DAC)*, 2025. [DOI](#)
- C2 A. Marchisio, E. Sychiucio, **M. Kashif**, and M. Shafique. “Cutting is all you need: Execution of large-scale quantum neural networks on limited-qubit devices.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. [DOI](#)
- C3 **M. Kashif**, S. Khalid, N. Innan, A. Marchisio, and M. Shafique. “Evaluating quantum amplitude estimation for pricing multi-asset basket options.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. [DOI](#)
- C4 **M. Kashif** and M. Shafique. “HQNET: Harnessing quantum noise for effective training of quantum neural networks in NISQ era.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. [DOI](#)
- C5 N. Innan, **M. Kashif**, et al. “Next-generation quantum neural networks: Enhancing efficiency, security, and privacy.” *IEEE International Symposium on On-Line Testing and Robust System Design (IOLTS)*, 2025. [Publisher](#)
- C6 T. Ahmed, A. Marchisio, **M. Kashif**, and M. Shafique. “Noisy HQNNs: A comprehensive analysis of noise robustness in hybrid quantum neural networks.” *International Joint Conference on Neural Networks (IJCNN)*, 2025. [DOI](#)
- C7 A. Basit, M. Shao, M. H. Asif, N. Innan, **M. Kashif**, et al. “PennyCoder: Efficient domain-specific LLMs for PennyLane-based quantum code generation.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2025. [DOI](#)
- C8 **M. Kashif** and M. Shafique. “Position Paper: Quantum neural networks—A paradigm shift in AI or a theoretical promise?.” *International Joint Conference on Neural Networks (IJCNN)*, 2025. [DOI](#)
- C9 A. Marchisio, M. U. Hafeez, N. Innan, **M. Kashif**, and M. Shafique. “Q-PORT: Quantum portfolio optimization with resource-efficient encoding and scalability analysis.” *International Conference on Quantum Engineering Sciences and Technologies for Industry and Services (QUEST-IS)*, 2025. [DOI](#)

- C10 N. Innan, **M. Kashif**, A. Marchisio, Y. S. Gan, F. Barbaresco, and M. Shafique. “QUAV: Quantum-assisted path planning and optimization for UAV navigation with obstacle avoidance.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. DOI
- C11 A. Basit, M. Shao, M. H. Asif, N. Innan, **M. Kashif**, et al. “QHACKBench: Benchmarking large language models for quantum code generation using PennyLane hackathon challenges.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. DOI
- C12 N. Innan, **M. Kashif**, et al. “Quantum amplitude estimation in practice: A case study in option pricing.” *International Conference on Quantum Engineering Sciences and Technologies for Industry and Services (QUEST-IS)*, 2025. DOI
- C13 J. Kaldari, **M. Kashif**, S. Al-Kuwari, and M. Shafique. “ResQGRNN: Quantum-compatible residual learning for graph recurrent neural networks.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. DOI
- C14 **M. Kashif** and M. Shafique. “NRQNN: The role of observable selection in noise-resilient quantum neural networks.” *CSCE 2024, revised selected papers, Communications in Computer and Information Science*, vol. 2257, Springer, 2025 (conference held 2024). DOI
- C15 K. Zaman, T. Ahmed, **M. Kashif**, et al. “Studying the impact of quantum-specific hyperparameters on hybrid quantum–classical neural networks.” *CSCE 2024, revised selected papers, Communications in Computer and Information Science*, vol. 2257, Springer, 2025 (conference held 2024). DOI
- C16 **M. Kashif**, et al. “Alleviating barren plateaus in parameterized quantum machine learning circuits: Investigating advanced parameter initialization strategies.” *Design, Automation & Test in Europe (DATE)*, 2024. DOI
- C17 **M. Kashif**, E. Sychiuco, and M. Shafique. “Investigating the effect of noise on the training performance of hybrid quantum neural networks.” *International Joint Conference on Neural Networks (IJCNN)*, 2024. DOI
- C18 K. Zaman, A. Marchisio, **M. Kashif**, and M. Shafique. “PO-QA: A framework for portfolio optimization using quantum algorithms.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024. DOI
- C19 **M. Kashif** and M. Shafique. “The dilemma of random parameter initialization and barren plateaus in variational quantum algorithms.” *IEEE International Conference on Rebooting Computing (ICRC)*, 2024. DOI
- C20 **M. Kashif** and S. Al-Kuwari. “Demonstrating quantum advantage in hybrid quantum neural networks for model capacity.” *IEEE International Conference on Rebooting Computing (ICRC)*, 2022. DOI
- C21 **M. Kashif** and S. Al-Kuwari. “Qiskit as a simulation platform for measurement-based quantum computation.” *IEEE International Conference on Software Architecture Companion (ICSA-C)*, 2022. DOI
- C22 M. Zahoor, F. Azam, M. W. Anwar, N. Yousaf, and **M. Kashif**. “A UML profile for the service discovery in the Enterprise Cloud Bus (ECB) framework.” *Complex, Intelligent, and Software Intensive Systems (CISIS)*, Springer, 2019. DOI
- C23 **M. Kashif**, I. Çiçek, and M. Imran. “A hardware efficient elliptic curve accelerator for FPGA based cryptographic applications.” *International Conference on Electrical and Electronics Engineering (ELECO)*, 2019. DOI
- C24 M. Rashid, M. W. Anwar, F. Azam, and **M. Kashif**. “Model-based requirements and properties specifications trends for early design verification of embedded systems.” *System of Systems Engineering Conference (SoSE)*, 2016. DOI

- C25 M. Imran, **M. Kashif**, and M. Rashid. “Hardware design and implementation of scalar multiplication in elliptic curve cryptography (ECC) over GF (2^{163}) on FPGA.” *International Conference on Information and Communication Technologies (ICICT)*, 2015. DOI

Preprints

- P1 **M. Kashif** and M. Shafique. “Beyond Logical Circuits: Hardware-aware analysis of expressibility and trainability in variational quantum algorithms.” *arXiv preprint*, 2026. [arXiv:2605.25552](#)
- P2 A. Marchisio, **M. Kashif**, N. Innan, and M. Shafique. “Hybrid quantum–classical neural architecture search.” *arXiv preprint*, 2026. [arXiv:2605.18345](#)
- P3 A. Marchisio, A. Ebrahim, N. Innan, **M. Kashif**, and M. Shafique. “QLIF-CAST: Quantum leaky-integrate-and-fire for time-series weather forecasting.” *arXiv preprint*, 2026. [arXiv:2605.18333](#)
- P4 R. Ahmed, **M. Kashif**, N. Innan, and M. Shafique. “Quantum vs. classical machine learning: A benchmark study for financial prediction.” *arXiv preprint*, 2026. [arXiv:2601.03802](#)
- P5 **M. Kashif** and M. Shafique. “Rethinking expressibility–trainability trade-off in hybrid quantum neural networks.” *arXiv preprint*, 2026. [arXiv:2605.25768](#)
- P6 J. R. M. Njiki, N. Innan, A. Marchisio, **M. Kashif**, J. M. Dricot, and M. Shafique. “Robustness evaluation of hybrid quantum neural networks under noise models via system-level error mitigation.” *arXiv preprint*, 2026. [arXiv:2604.17515](#)
- P7 M. Atallah, N. Innan, **M. Kashif**, and M. Shafique. “Investigating different barren plateaus mitigation strategies in variational quantum eigensolver.” *arXiv preprint*, 2025. [arXiv:2512.11171](#)